Chemical Reactor

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1 Introduction

The total energy of the system is given as

$$E_{tot} = E_{kin} + E_{rot} + E_{vib} \tag{1.1}$$

the individual components can be expressed as assuming a common kinetic temperature of the mixture

$$E_{kin} = \frac{3}{2} k_B T \sum_i N_i \tag{1.2}$$

$$E_{rot} = k_B \sum_i^{N_s} N_i \frac{\xi_i}{2} T_i \tag{1.3}$$

$$E_{vib} = k_B \sum_i^{N_s} \left(N_i \sum_m g_{i,m} \frac{\theta_{i,m}}{\exp(\theta_{i,m}/T_{i,m}) - 1} \right) \tag{1.4}$$

where the index i denotes the species and m the vibrational mode. We assume a constant volume thus we can work with per volume quantities and replace the particle number N with number density n. Assuming a thermalized rotational temperature we can write the per particle energy as

$$e_{kr,i} = \frac{3 + \xi_i}{2} k_B T \tag{1.5}$$

$$e_{v,i,m} = k_B g_{m,i} \frac{\theta_{i,m}}{\exp(\theta_{i,m}/T_{i,m}) - 1} \tag{1.6}$$

and therefore

$$\underbrace{E_{kin} + E_{rot}}_{E_{kr}} = \sum_i n_i e_{kr,i} = T k_B \sum_i n_i \frac{3 + \xi_i}{2} \tag{1.7}$$

$$E_{vib} = \sum_i \left(n_i \sum_m e_{v,i,m} \right) \tag{1.8}$$

We have

$$\frac{\partial E_{kr}}{\partial t} = \sum_i \left(\frac{\partial n_i}{\partial t} e_{kr,i} + n_i \frac{\partial e_{kr,i}}{\partial t} \right) + \Delta E_R \tag{1.9}$$

$$\frac{\partial E_{vib}}{\partial t} = \sum_i \sum_m \left(\frac{\partial n_i}{\partial t} e_{v,i,m} + n_i \frac{\partial e_{v,i,m}}{\partial t} \right) \tag{1.10}$$

$$\Delta E_R = \sum_r \nu_r \Delta e_r \tag{1.11}$$

$$\frac{\partial n_i}{\partial t} = \sum_k n_i n_k r_{i,k}(T) \tag{1.12}$$

$$\frac{\partial e_{kr,i}}{\partial t} = -\sum_m \frac{\partial e_{v,i,m}}{\partial t} \tag{1.13}$$

$$\frac{\partial e_{v,i,m}}{\partial t} = \frac{e_{v,i,m}(T) - e_{v,i,m}(T_{i,m})}{\tau_{i,m}} \tag{1.14}$$

2 Fundamentals

Given a closed system of constant volume the total energy of the system E is given as

$$E = \sum_{s=1}^{N_s} e_s n_s \tag{2.1}$$

where e_s is the energy of a particle of species s and n_s is the number density of said species. The total energy has the unit $[E] = \mathrm{J\,m^{-3}}$. The energy of a species consists of the kinetic energy e_k, the rotational energy e_r and the vibrational energy e_v. It is assumed that the translational energy is in equilibrium between the individual species $e_{t,s} = e_t$ therefore we get

$$E = e_t \sum_{s=1}^{N_s} n_s + \sum_{s=1}^{N_s} (e_{r,s} + e_{v,s}) n_s. \tag{2.2}$$

For polyatomic species the vibrational energy is the sum of vibrational energies of individual modes m

$$E = e_t \sum_{s=1}^{N_s} n_s + \sum_{s=1}^{N_s} \underbrace{\left(e_{r,s} + \sum_{m=1}^{N_m} e_{v,s,m} \right)}_{e_{int,s}} n_s. \tag{2.3}$$

Assuming chemical reactions the change of energy is

$$\frac{\partial E}{\partial t} = \dot{r} \tag{2.4}$$

where $\dot{r}$ is the energy change due to chemical reactions

$$\dot{r} = \sum_{r=1}^{N_r} \nu_r \Delta e_r \tag{2.5}$$

where ν_r is the rate of reaction r and Δe_r is the energy release of said reaction. The reaction rate is give as $[\nu_r] = \mathrm{reaction\,s^{-1}\,m^{-3}}$ and $[\Delta e_r] = \mathrm{J}$. The total change of the energy in time is

$$\frac{\partial E}{\partial t} = \frac{\partial e_t}{\partial t} \sum_{s=1}^{N_s} n_s + e_t \sum_{s=1}^{N_s} \frac{\partial n_s}{\partial t} + \sum_{s=1}^{N_s} \frac{\partial e_{int,s}}{\partial t} n_s + \sum_{s=1}^{N_s} e_{int,m} \frac{\partial n_s}{\partial t} \tag{2.6}$$

$$= \frac{\partial e_t}{\partial t} \sum_{s=1}^{N_s} n_s + \sum_{s=1}^{N_s} \frac{\partial e_{int,s}}{\partial t} n_s + \sum_{s=1}^{N_s} e_s \frac{\partial n_s}{\partial t}. \tag{2.7}$$

The change of number density in time is given as

$$\frac{\partial n_s}{\partial t} = \sum_{r=1}^{N_r} \nu_r \alpha_{r,s} \tag{2.8}$$

where $\alpha_{r,s}$ is the stoichiometric coefficient of species s in reaction r.

The change of an internal energy mode being it rotational or vibrational is given as

$$\frac{\partial e_m}{\partial t} = \frac{e_m(T) - e_m}{\tau_m} \tag{2.9}$$

where τ_m is the relaxation time of mode m and T is the translational temperature given as

$$e_t = \frac{3}{2} k_B T \tag{2.10}$$

i.e.

$$T = \frac{2}{3} \frac{e_t}{k_B}. \tag{2.11}$$

The change of tranlational energy is therefore given as

$$\frac{\partial e_t}{\partial t} = -\frac{\sum_{s=1}^{N_s} \partial_t e_{int,s} n_s + \sum_{s=1}^{N_s} e_s \partial_t n_s}{\sum_{s=1}^{N_s} n_s} \tag{2.12}$$

Using $n_{tot} = \sum_{s=1}^{N_s} n_s$ we can write

$$\frac{\partial e_t}{\partial t} = -\frac{1}{n_{tot}} \sum_{s=1}^{N_s} \left(\frac{\partial e_{int,s}}{\partial t} n_s + e_s \frac{\partial n_s}{\partial t} \right). \tag{2.13}$$